

```

motion service if needed
$ sudo systemctl restart motion # Restart
the motion service
$ sudo systemctl status motion # Check
the status of the motion service
$ sudo nano /etc/motion/motion.conf #
Open the motion configuration file; most
settings are preconfigured and ready
to use

```

Most settings are preconfigured and ready to use. No changes are required unless the exact modifications are known. Missing files or directories such as motion.log or process_id_file may occasionally occur. In such cases, the necessary directories or files should be created, appropriate permissions applied, and the service restarted.

The Motion software is highly flexible, and all options are configurable within the motion.conf file. The file should be read carefully to understand each option, for example, how to change the log file location, the port for streaming, and other parameters. For ease of use, a sample motion.conf file has been included as an attachment.

ZeroTier. Fig. 2 illustrates how ZeroTier enables the creation of a virtual private network (VPN) tunnel across the internet, allowing multiple devices in different locations to communicate securely as if they were on the same local network. The key components shown include various devices labelled D1 through D5, internet access points (clouds), a shared hotspot for local devices, and the ZeroTier (ZT) engine, which facilitates encrypted communication.

In this setup, devices D1, D2, and D3 are connected to a common hotspot that provides internet access. These devices use ZeroTier to connect the virtual network via the public internet. Devices D4 and D5, located in different remote locations and connected to the internet through separate networks, also install ZeroTier and join the same virtual network. This configuration

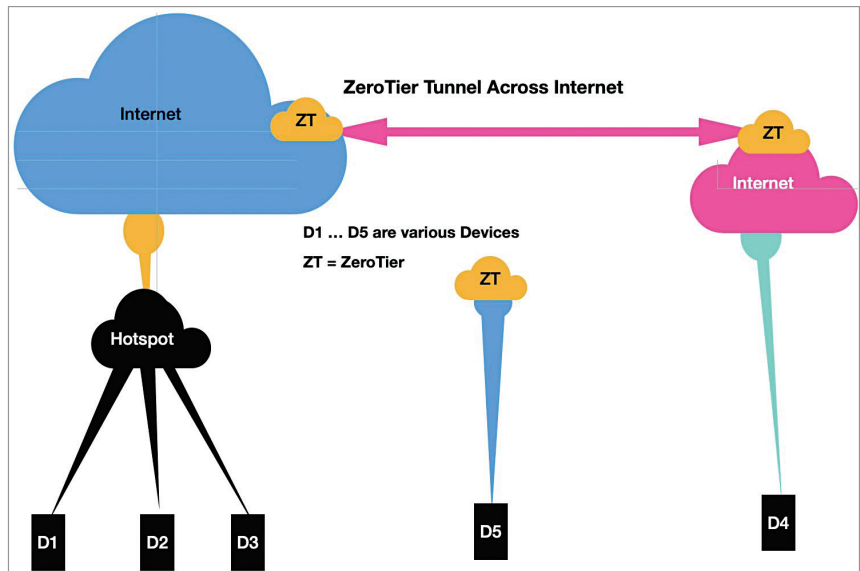


Fig. 2: Illustration of ZeroTier

Create A Network

Your Networks

Networks: **2 / 3**

Included Devices: **0 / 10**

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SEARCH

NETWORK ID	NAME	DESCRIPTION	SUBNET	NODES	CREATED
51279c...92d1	berapi-2		172.24.0.0/16	5	2024-10-23
80f...	.6ec berapi		192.168.196.0/24	2	2024-10-21

Fig. 3: ZeroTier central dashboard

ensures that all devices are logically connected, regardless of their physical location.

ZeroTier creates a secure tunnel across the internet, allowing peer-to-peer or virtual LAN-like communication among all connected devices (D1-D5). This approach is suitable for scenarios such as remote device access, secure communication without traditional VPNs, and building a private network across different sites.

This virtual networking tool enables secure, encrypted one-to-one communication across the internet without complex configurations such as port forwarding. It creates a secure tunnel for the video stream to be transmitted and

accessed by authorised devices anywhere in the world. Once the concept is clear, the process is straightforward.

With Motion running on the Raspberry Pi Zero, the stream is accessible on port 8081 within the local network. Any device on the same Wi-Fi network can view it by entering:

```
http://<pi-zero-ip>:8081
```

To make the stream available outside the local network (beyond the mobile hotspot's NAT firewall), ZeroTier is required to create a secure pipeline between devices. When connected to the internet, ZeroTier establishes a virtual tunnel through which devices D1 through D5 connect, as shown in Fig. 2.